

PRINTING IDS SCREENS

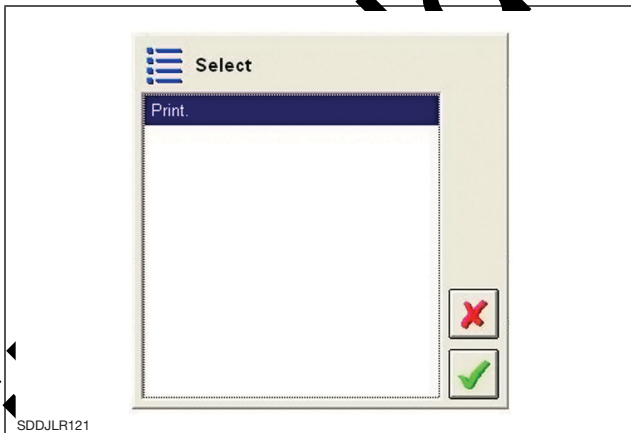
IDS includes a program called PDF995, which allows the user to 'print' screen images and save them as a Portable Document Format (PDF) file. These files can then be saved to a preferred file destination on the computer, or saved to the USB drive and transferred to another computer for printing, e-mailing, etc. When a problem occurs on IDS, it is most helpful if the screen images are printed and sent to the Support Desk.

To print a screen image, select the desired screen, then select the Select System Options function button.

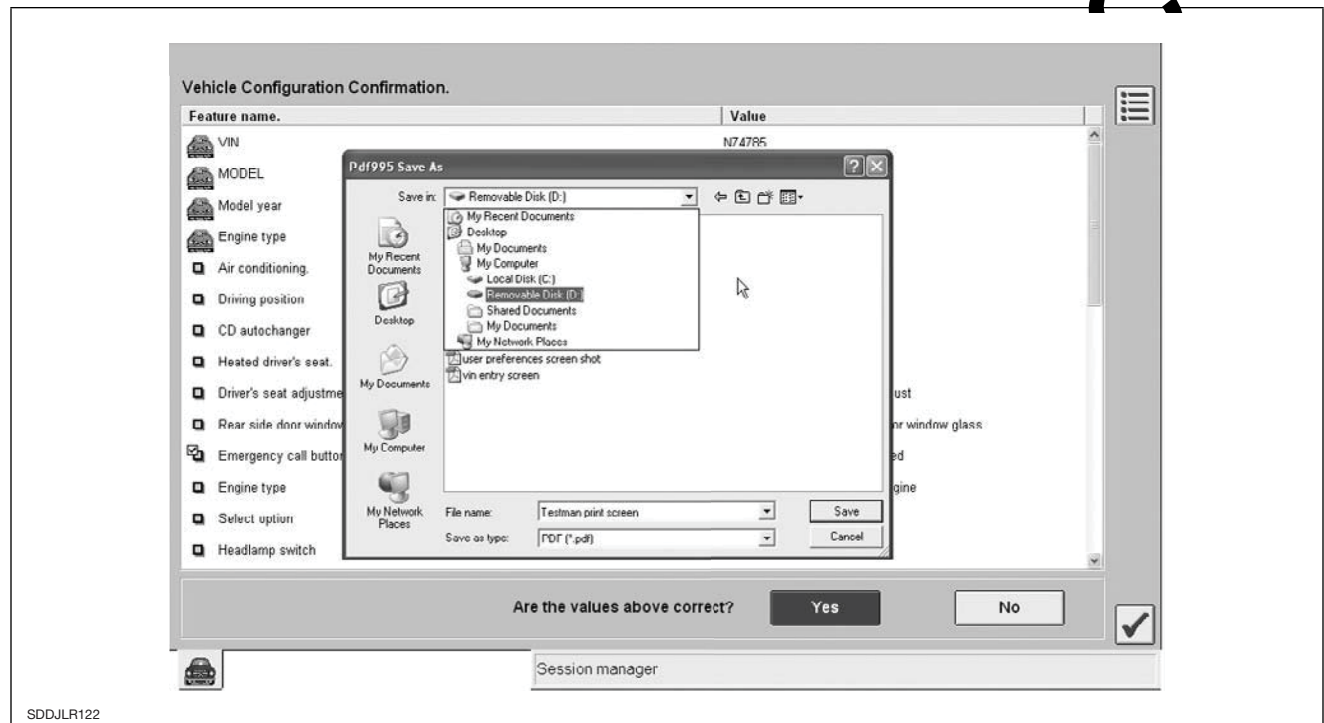
Select System Options Function Button



A dialog box will appear, with option 'Print' or 'Print screen' highlighted.



Select the confirmation 'tick' to print/save the selected screen. A 'Pdf995 Save As' window will be displayed.

'Pdf995 Save As' Pop-Up Screen

Select a location where the file is to be saved and give the file a name.

NOTE: IDS will assign a default file name, but it is recommended that you assign a specific name that will help you identify and locate the file later (for example, 'VIN_012345_DTC1').

Save to USB Drive

To save the file to the USB drive (make sure the drive is connected), select 'Removable Drive' from the list, then select 'Save.' The file is now saved to the USB drive.

Adobe Reader will open, displaying an image of the screen you have printed. Close the document.

Save to Computer Hard Drive

To save the file to the IDS laptop's hard drive, select an appropriate location (for example 'Desktop' --> 'My Documents'), then select 'Save.' The file is now saved to the hard drive.

Again, Adobe Reader will be open displaying an image of the screen you have captured. Close the document.

To locate the file, open the desktop and go to the location where the document was saved. The file can be copied to the USB drive and transferred to another computer or printer if required.

SDD MONITOR TRACE

The SDD Monitor is a powerful data collection tool that enables users to capture details of a failed SDD process. When Jaguar / Land Rover engineers must diagnose a failed procedure, the SDD Monitor information is the only way to effectively identify the root cause.

Example: Monitor trace provides additional details to troubleshoot a failed procedure, like module replacement or key programming. Information from the Monitor will allow JLR to determine if the failed procedure is within the vehicle or SDD itself.

Before exiting from the diagnostic routine, select the 'Alt' + 'Tab' keys to display the Main Menu. Select 'SDD Monitor' from the 'SDD Configuration' menu.

SDD Monitor will open and create a trace of all data relating to the diagnostic routine performed. Name and save the file to a preferred destination, where it can be retrieved later. It is recommended that the file name contains the VIN, since this will assist in referring to it easily. The file can be transferred to another computer using the USB drive and, if required, can be sent to engineering for analysis.

The trace file will show all details of the diagnostic routine performed, including the VIN, test performed, which answers had been selected from the pop-up messages, etc.

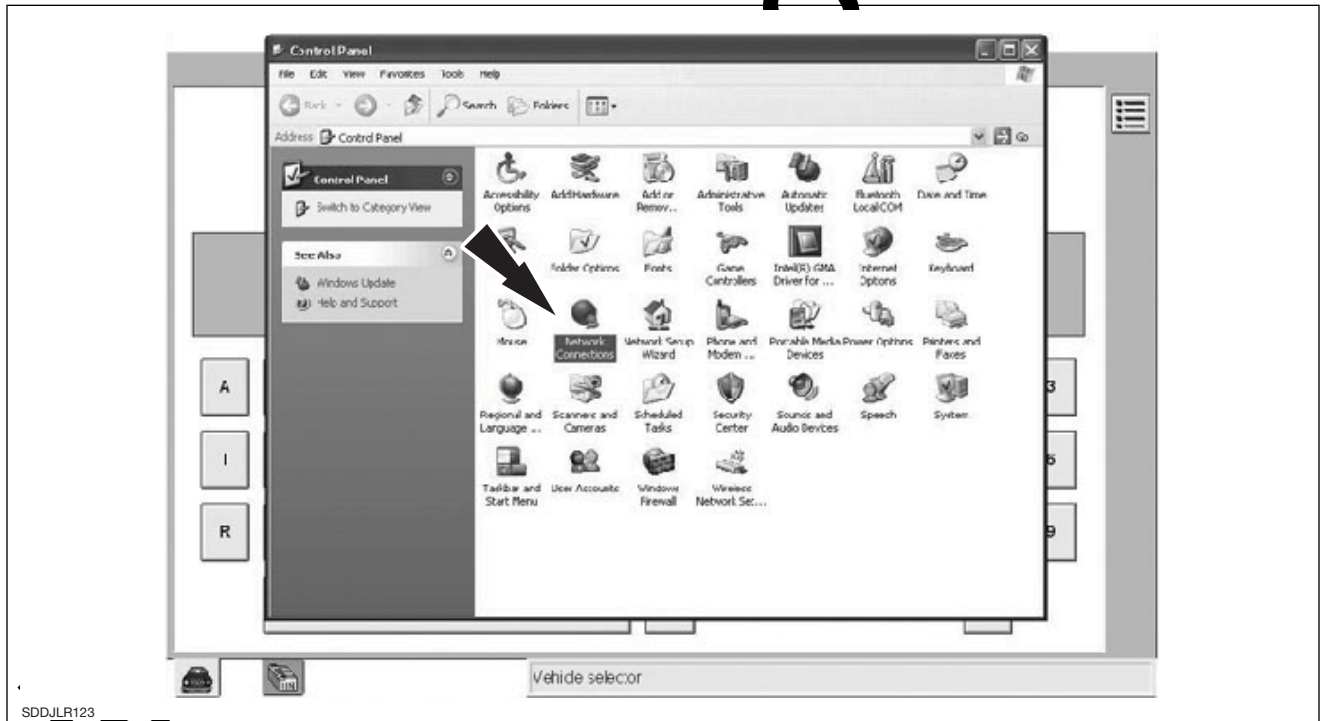
NOTE: Due to frequent updates to SDD Monitor operation, please review the latest Special Service Messages (SSMs) to set up and run a Monitor Trace effectively.

DISABLING WINDOWS XP NETWORK WARNINGS

The network established between the VCM and the IDS laptop is considered a 'limited' network by the Windows environment. Because of this, Windows will issue a warning whenever the VCM connection is established or terminated. The warnings are displayed in balloon format on the bottom right of the screen, and may cover some function buttons.

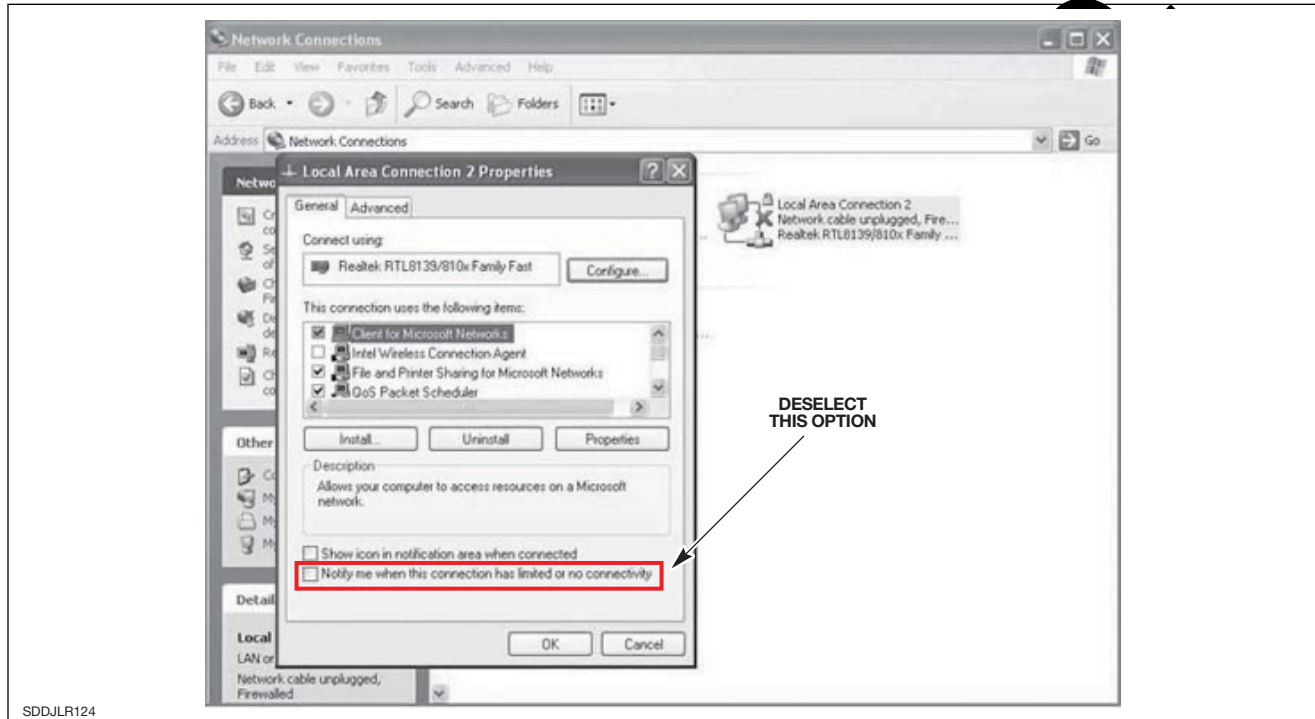
These warnings, and warnings which occur whenever power to the VCM is interrupted, can be disabled within the Windows Control Panel as follows:

- Select 'Network Connections' from the Control Panel.



SDDJLR123

- Right-click on 'Local Area Connection 2' and select 'Properties'
- Deselect the bottom box, labeled 'Notify me when this connection has limited or no connectivity' and click 'OK'



This process will need to be repeated for each of the IDS network connections.

NOTE: Do not turn off notification for the wireless network.

IDS-SDD-JLR: IDS Symptom Driven Diagnostics



Maintenance and Support



www.jlridssddmongoose.com

This publication is intended for instructional purposes only. Always refer to the appropriate service publication for specific details and procedures.

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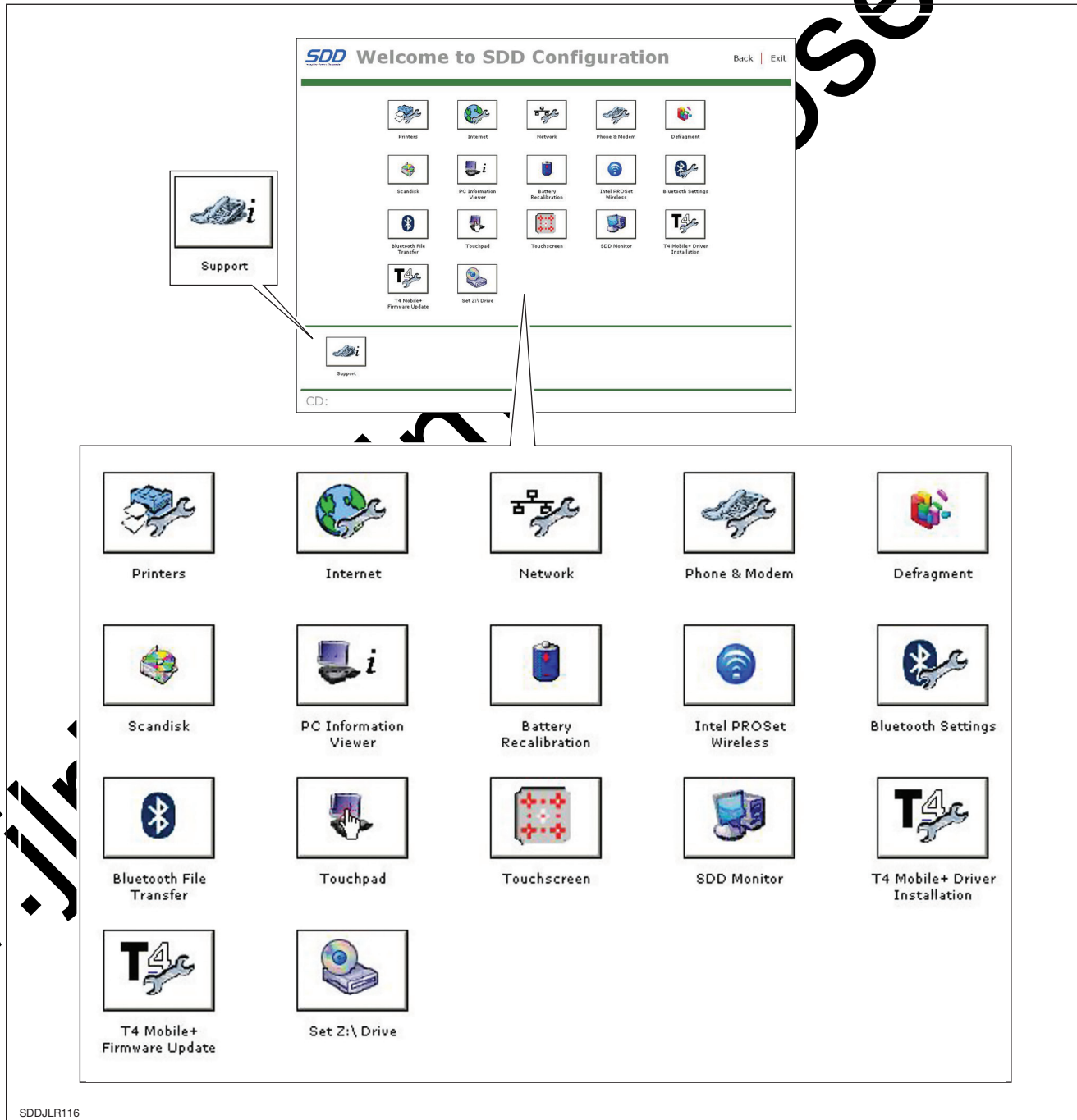
<https://www.jlridssddmongoose.com/sdd-symptom-driven-diagnostics/>

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www.jlrldsddmongoose.com

IDS MAINTENANCE

Several Maintenance features are available to keep IDS running in optimal condition. Maintenance features are accessed from the SDD Configuration menu.



Scandisk

A scandisk should be performed once a month. This action checks the drive for corrupt files or bad segments and repairs them as required.

Disk Defragmenter

Because of the way a computer stores data on its hard disk, the storage space can become 'fragmented' over time, with bits of data for a single operation stored in pieces around the hard drive. Once this happens, the computer has to search simultaneously in several areas of the hard drive to locate necessary data, which can slow down the operation of the computer. Disk Defragmenter reassembles fragmented data (the operation is called 'defragging') so that the hard drive can work more efficiently. The Disk Defragmenter is accessed by selecting 'Defragment' from the SDD Configuration menu.

It is recommended that the defragmenter is used at a convenient time following the installation of new diagnostic software. Defragging the hard drive may take some time to complete, so only carry out this procedure at the end of a working day when IDS will not be required. Monthly defragging is recommended by JLR.

Touch Screen Display Calibration

If the accuracy of the TSD has depreciated, it can be recalibrated by selecting 'Touchscreen/Touchpad' from the SDD Configuration menu. The mouse properties will be displayed. Select the 'Touch screen' tab and then 'calibration'. The calibration screen will be displayed.

Using the stylus, touch and hold the '+' symbol for approximately one second. The symbol will then move to another position where the procedure will be repeated. Repeat the process for each position of the symbol. Once all nine '+' positions have been verified, complete the calibration by selecting 'Enter'.

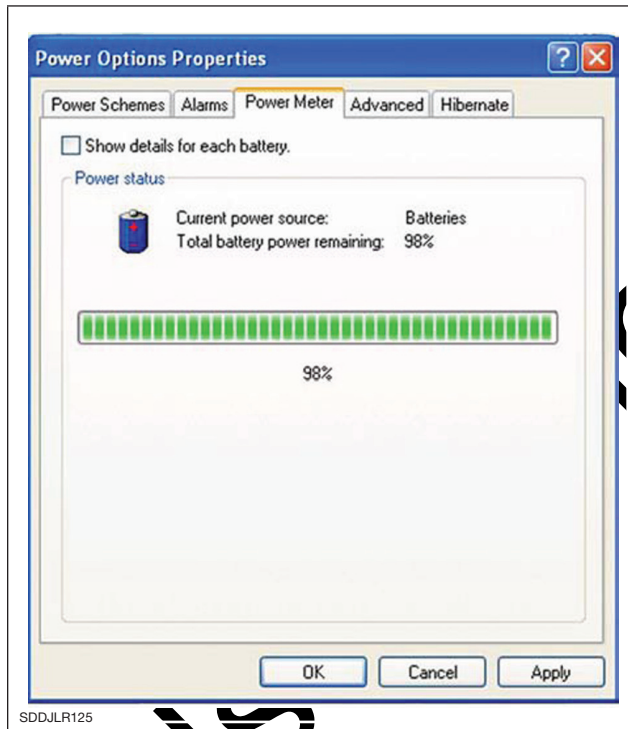
Calibration Screen



Laptop Battery

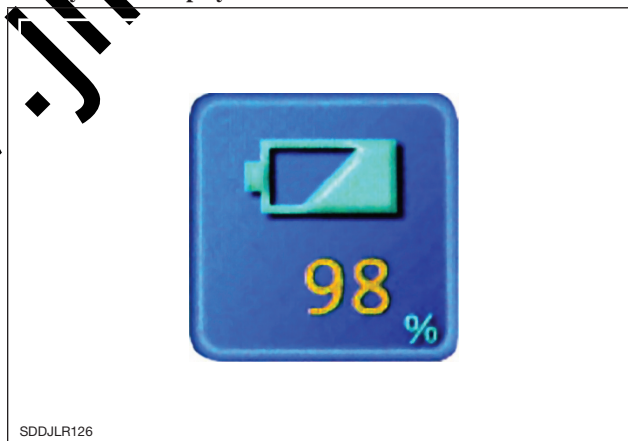
Battery Power Status

The condition of the internal battery can be viewed in the 'Power Options Properties' window by selecting the 'Power Meter' tab. The current power source and the percentage of battery power remaining will be displayed.



The battery condition can also be displayed by selecting 'Fn' + 'F9' on the keyboard.

Battery Power Display



Battery Calibration

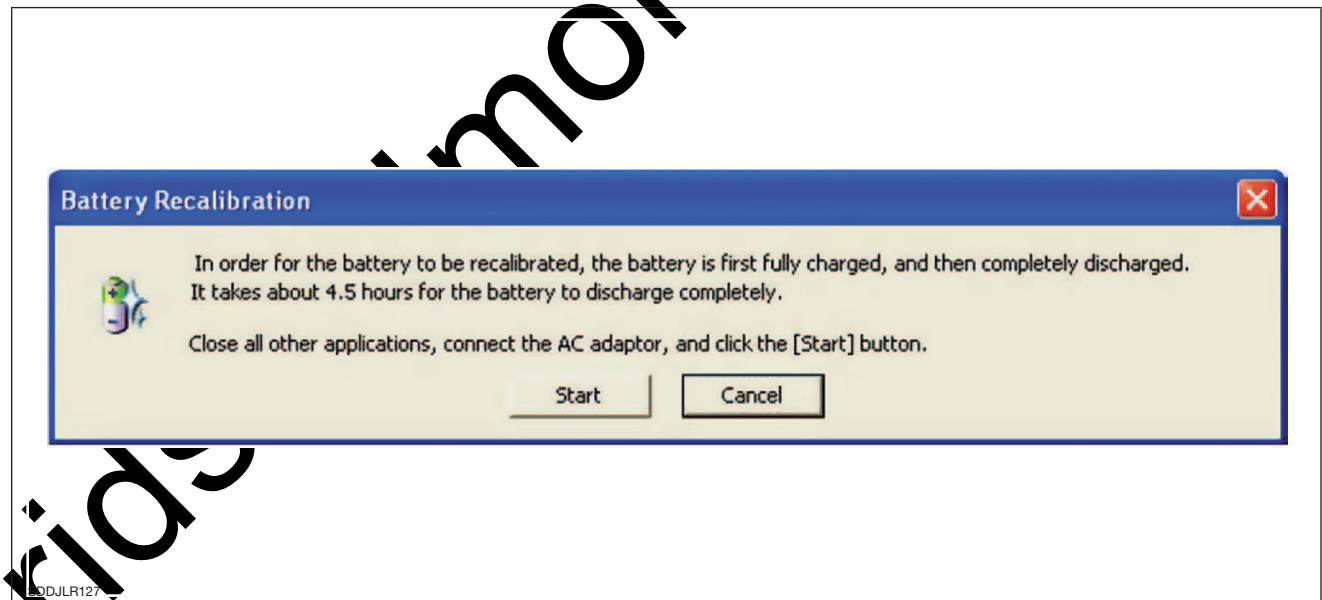
As the battery deteriorates the remaining battery capacity may not be displayed accurately. In this situation, select 'Battery Recalibration' from the SDD Configuration menu to recalibrate the battery. The battery will be charged (if not already fully charged), then fully discharged.

During battery recalibration, the following actions are carried out:

- Battery will be charged (providing it is not already fully charged)
- Battery will be discharged
- Computer will shut down
- Battery will be charged

NOTE: The calibration procedure also recalibrates the battery power meter to ensure its accuracy.

It is recommended that battery recalibration is carried out monthly. Battery recalibration can take up to 14 hours to complete and so should be carried out overnight when IDS is not required. Battery recalibration can only be carried out when IDS is connected to an AC power supply.



IDS SUPPORT

Contacting the Support Desk

In the event that you experience any problems with using IDS and you are unable to resolve them by repeating the procedure or by consulting IDS literature, contact the IDS support desk for assistance.

To contact IDS Support:

- In the U.S. and Canada, call **(866) 628-5508**
- In Mexico, call **(800) 504-5330**
- From TOPIx, go to **www.spxtools.com**

The support desk will carry out a call qualification check, making sure that your dealer code is valid and that your support contract is up to date.

NOTE: IDS support is provided by SPX (the IDS supplier), not by the Dealer Technical Support Hotline. The Dealer Technical Support Hotline only provides support for vehicle concerns.

Before contacting either of the support teams, make sure you have the following details available:

- Dealer Name
- Dealer code
- Telephone Number
- Fax Number
- Contact name
- IDS Serial/Model Number
- VIN number of the vehicle being tested
- Details of the test being carried out
- The details and description of the fault
- Details of any error messages displayed
- The IDS software application details.
 - **Example:** IDS software release number 2

Details of IDS Test Function Being Used

This information is vital to an IDS engineer who is trying to reproduce and solve the problem.

In addition to the Meritor 2k2 data, vehicle information also should be included with the exact vehicle model, derivative, model year and VIN, plus any component numbers or codes relevant to the area of the vehicle being worked on. When combined with details of the reported fault that you are trying to diagnose, this may give IDS engineers a short cut to a solution based on previous experience.

Some diagnostics may be reached using several different routes. It is vital that the IDS engineers follow exactly the same diagnostic route as the technician experiencing the problem. It would help if you made a note of each screen where you made a decision, so the engineer can follow exactly the same route.

IDS Example:

- Select Configuration tab
- Select Configure new modules
- Select Datalogger tab
- Select Global options

T4 Example:

- Select Diagnostic System
- Select Security
- Select Locking/Unlocking
- Select guided diagnostics

Continue the route until you reach the screen where the fault occurred.

In addition, it is helpful to IDS engineers if they are informed of difficulties experienced when using IDS even though you may have resolved the problem yourself. This information can then be used to prevent problems occurring in the future.

SMS (SPX Global Support Management System)

Once a call has been made to the IDS support desk, details will be logged by the desk on the SMS Global reporting system. The dealer can log onto the site and track the progress of the complaint/fault. All registered dealers will be able to log onto the site and check the progress of the help desk regarding their complaint/fault. Dealers will not be able to view details of calls made by other dealers.

To access the SMS site online, go to: <http://sms.spx.com>

SMS Login

SMS
SPX Global Support Management System

SPX
SERVICE SOLUTIONS

SPX GLOBAL SUPPORT MANAGEMENT SYSTEM

- Online tracking of support requests
- Find Asset Configurations and Projected Install Dates
- Look up technical manuals and site install documents
- Track repair status and details
- Personalized view and access

SMS is the latest initiative from SPX Service Solutions in its continuous commitment to provide world class quality and support.

In the increasing complexity of service diagnostics and information systems, review all aspects of the system and be informed. This support infrastructure will ensure fast turn around on customer requests and provide up to date information.

Login >

Email

Password **Log In**

☐ Save Email on this computer

Forgot Password? Click Here!!
New Users Register Here!!

[Contact Us](#)

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SDDJLR133

The dealer must first register on the site to gain access. Once access has been given, dealers will be able to view details of a current complaint, and also historic problems incurred by the dealer. Comments and corrective action details will be entered on the site by the help desk.

Dealers will be able to view the following information:

- Date and time help desk was contacted
- Contact details of dealership
- Nature of the complaint
- Action taken by help desk
- If the complaint is current or closed

www.jlridssddmongoose.com

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IDS-SDD-JLR: IDS Symptom Driven Diagnostics



Datalogger



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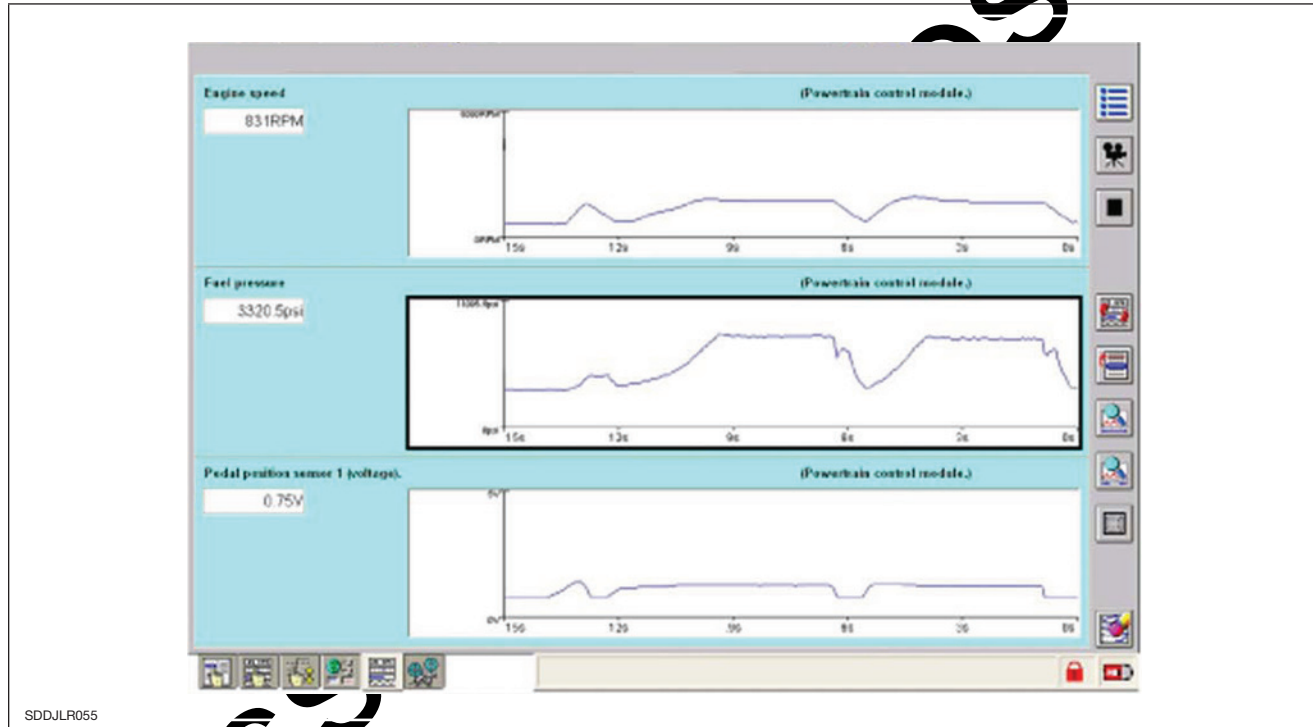
<https://www.jlridssddmongoose.com/sdd-symptom-driven-diagnostics/>

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OVERVIEW

Datalogger acquires, displays and records vehicle signals. It also permits the value of output signals from a control module to be controlled. The user can decide which signals are to be monitored, how they are formatted, sampled and scaled for display. Limits and trigger settings can also be set.

Datalogger Live Display Screen

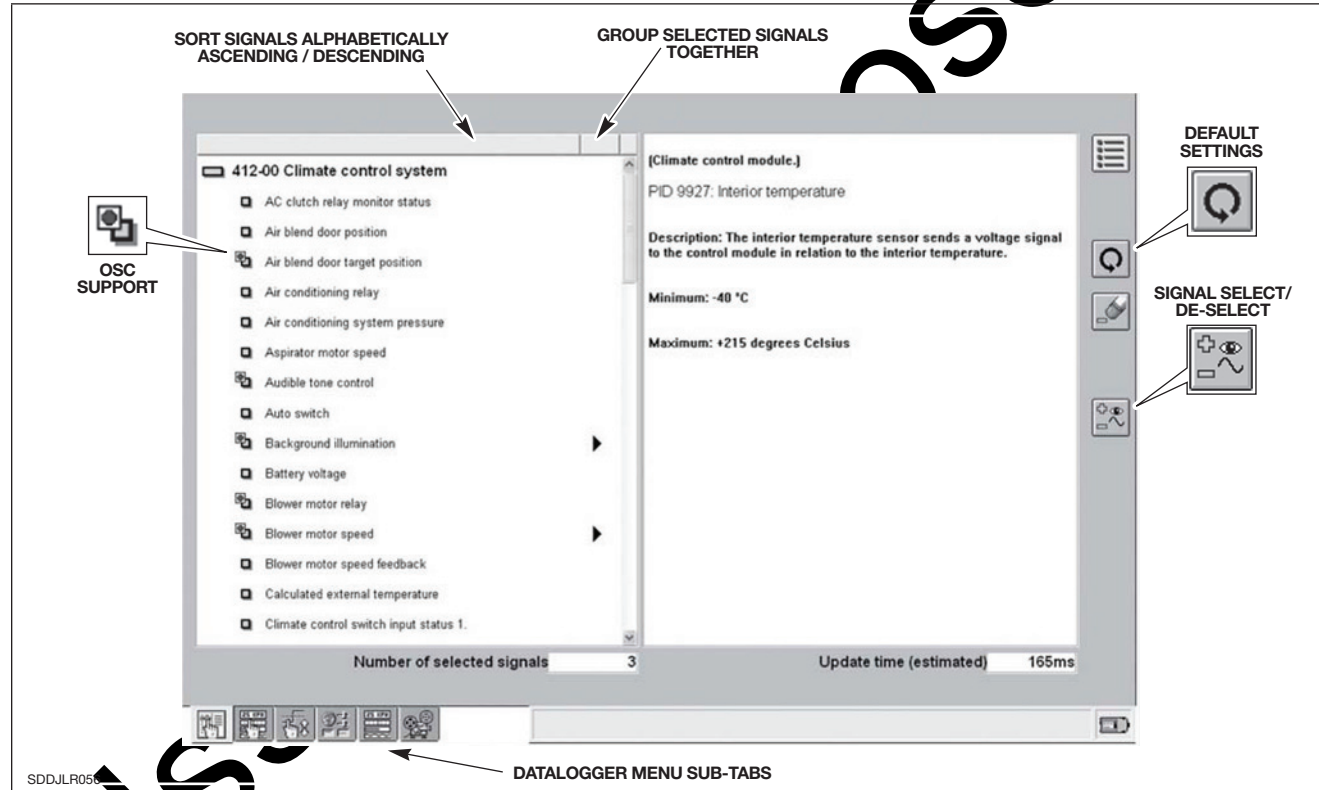


Prior to selecting Datalogger, the user should select the desired system for Datalogger monitoring using the Content Model; the first level of the Content Model is recommended for optimal results.

Datalogger Signal Selection

The Signal Selection sub-tab menu is displayed when Datalogger is first opened; this is the default. This provides an opportunity to immediately select or de-select any signals. To select a different signal while working in other Datalogger menus, the user will return to the Signal Selection menu by selecting the sub-tab.

Datalogger Signal Selection Menu Screen



Signal Selection Screen Layout

The left-hand panel displays a list of all available signals. The right-hand panel is used to display information about selected signals, such as description, standard units, operating range, etc.

The signal menu uses a hierarchical structure, similar to the menu structure of the Content Model. By selecting the '+' or '-' icon adjacent to a system, the menu will be either expanded or collapsed. Signals can be selected from anywhere within the menu structure.

If the vehicle system was selected from the Content Model prior to launching Datalogger, the Output State Control (OSC) function (where applicable) will be configured. Components that are supported by OSC are indicated by an enhanced icon adjacent to the signal.

Highlighting a signal in the menu will display additional information about the signal in the right-hand pane; it does not select the signal for logging. To select a signal for display or capture, highlight it and select the Signal Select / De-Select Function Button. The number of signals selected is always displayed at the bottom of the screen.

Sorting and Grouping Signals

The signals can be displayed in ascending or descending alphabetical order; sort the signals by clicking on the wide bar above the list.

The selected signals can be grouped together by clicking on the narrow bar to the right of the alphabetical sort bar. Grouping the signals serves two purposes:

- It allows the user to easily view the signals that have been selected
- It allows all selected signals to be de-selected at the same time, rather than individually.

USING DATALOGGER

Datalogger Sub-Tabs

Once the desired vehicle signals have been selected, use the Datalogger sub-tabs to switch between Datalogger's various functions.

Signal Selection

Select the signals for capture.

Signal Configuration

Set signal attributes including Display, OSC, and Analog.

Trigger Configuration

Set the triggering limits of a selected signal.

Global Configuration

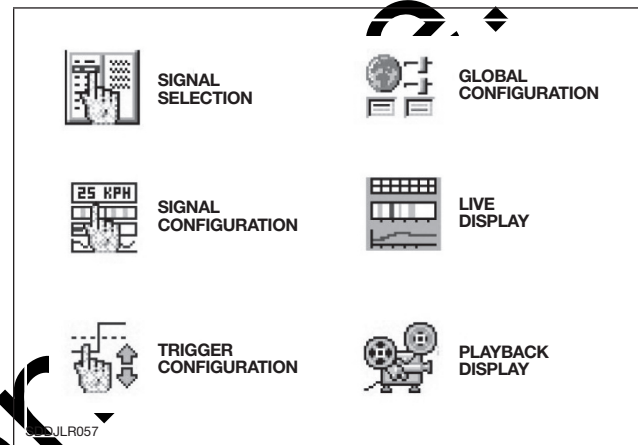
Set the parameters of all selected signals and display a summary of information about them.

Live Display

Capture selected signals and display them in real time.

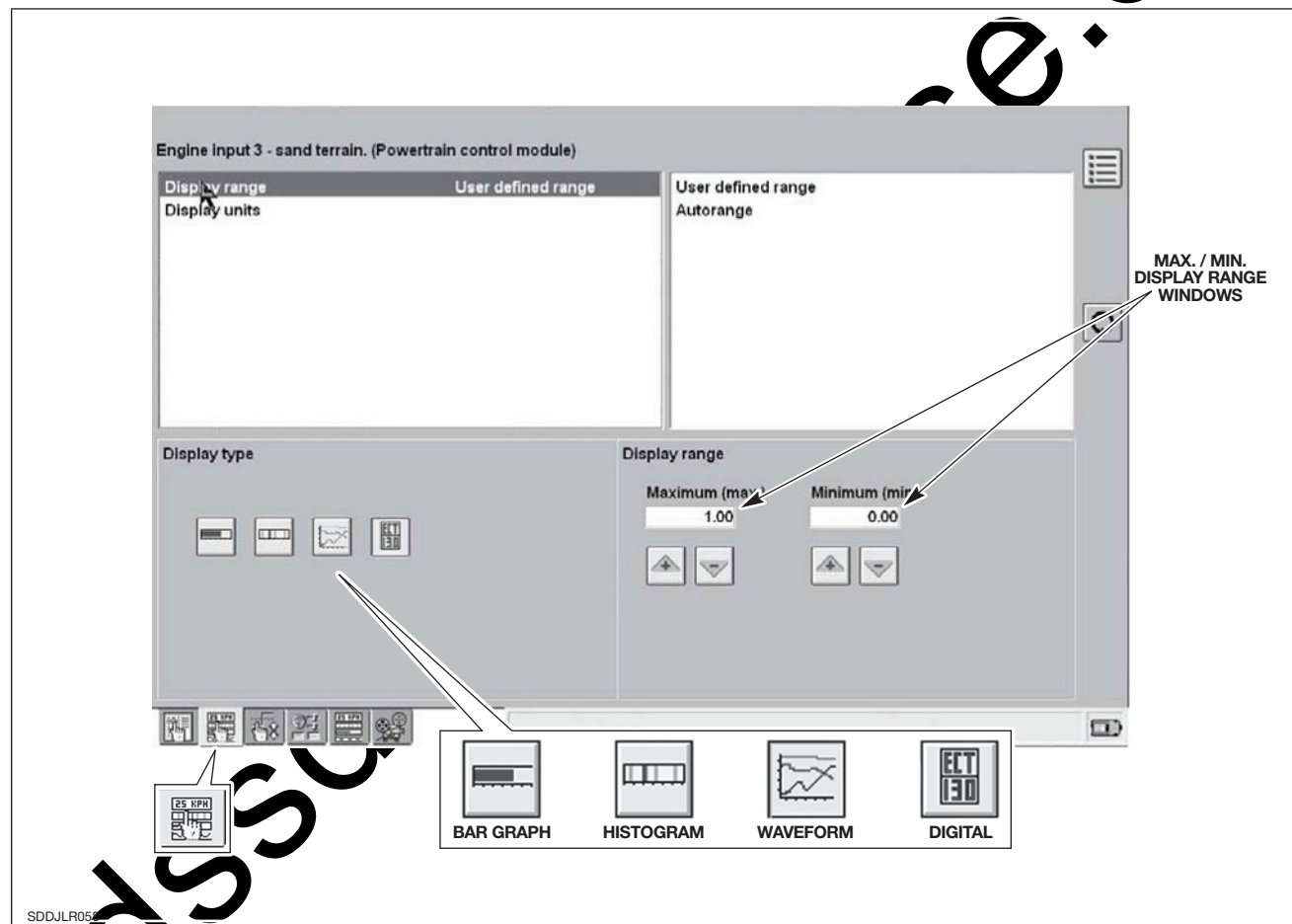
Playback Display

Recall and display previously captured signal data.



Datalogger Signal Display

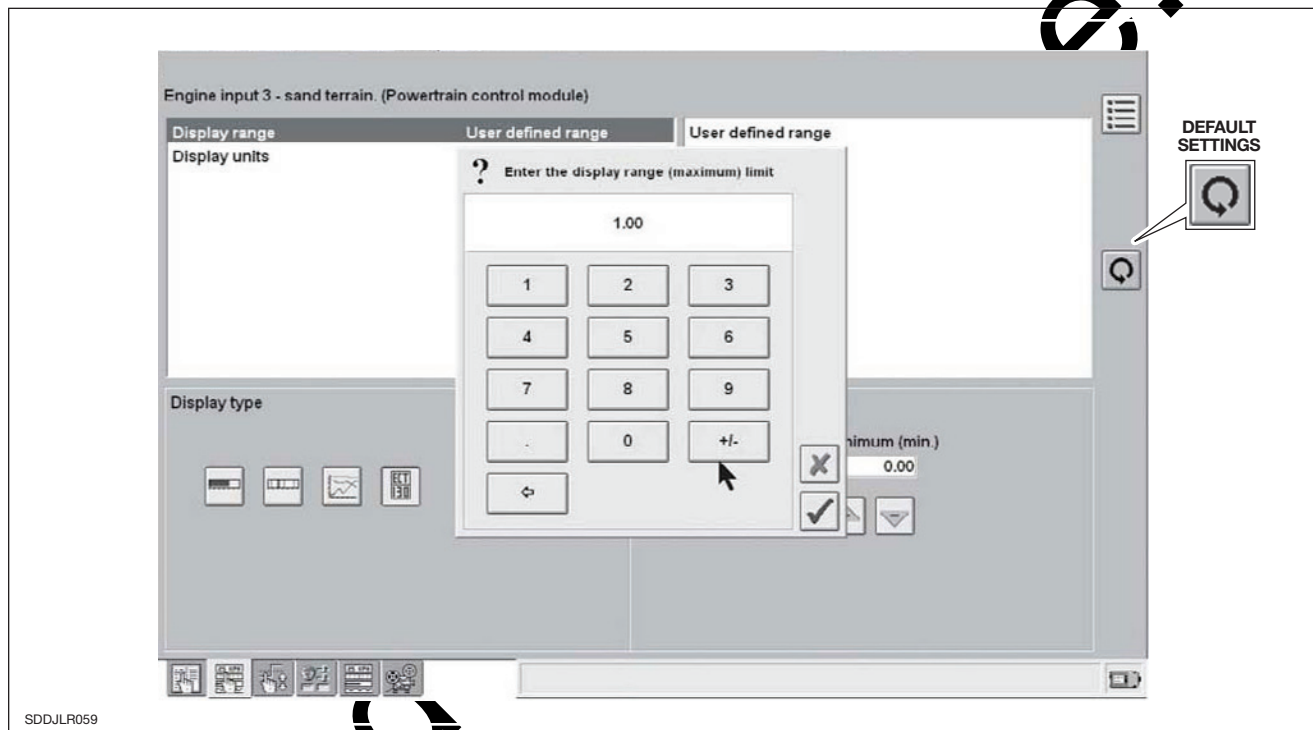
The Signal Configuration sub-tab allows the user to select the display type and to adjust the display range.



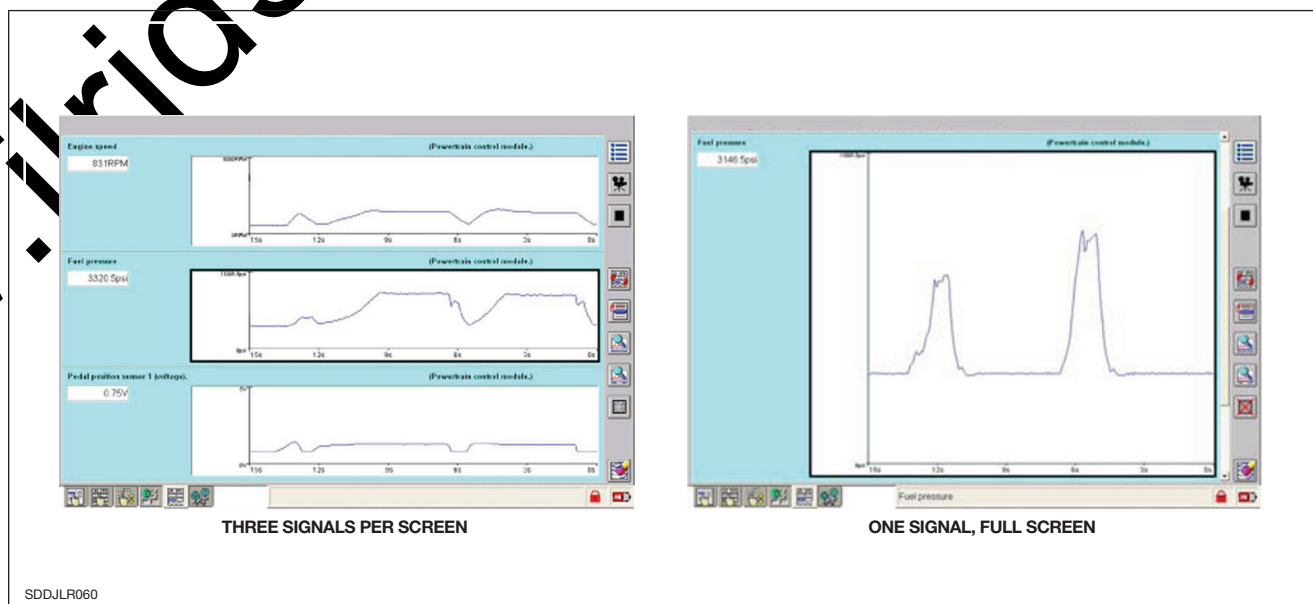
The 'Display type' buttons allow the user to change the form in which Datalogger signals are displayed, while 'Display Range' allows the user to modify the scale of the signal displayed.

For example, the fuel pressure Autorange may display the pressure on a scale of 1 to 100 psi; by highlighting 'User Defined Range' the user can set the minimum/maximum range of the display using the increment buttons from 1 to 50 psi, which allows the user to read the signal more precisely.

A numerical keypad is available to quickly change the value by a large amount. 'Tick' the TSD in the Min. or Max. display range window to display the keypad. The values can be returned to the default setting quickly by selecting the Default Settings function button.



The order in which the signals are displayed initially depends on the order in which they were selected. However, the order in which they are displayed can be changed. Up to three signals can be displayed on each screen, or one signal can be expanded to the full screen.

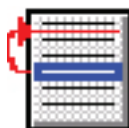


Changing Signal Display Order

To change the display order of the signals:

1. Select the signal you wish to move.
2. Selecting the Move Signal Position function button.
3. Select the signal that occupies the point where you wish the selected signal to be moved to.

Move Signal Position Function Button



Expand Signal Display to Full Screen

To expand a signal, select the signal and operate the Full Screen function button. To return the signal display to normal view, operate the Full Screen button again.

Full Screen Function Button

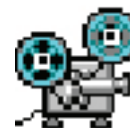


Capture Playback

A Datalogger signal can be captured and viewed later for analysis. The signal can be captured manually by selecting the 'Trigger Datalogger Recording' button, or a trigger can be set so that when the selected trigger conditions are met, the Datalogger trace will automatically be recorded.

To view a captured signal, select the Playback Viewer sub-tab.

Playback Viewer Sub-Tab



Select the sub-tab to open the Playback screen. The format mirrors the Live Display screen: the range and display formats can be modified, and the X-axis can be expanded and contracted using the appropriate buttons (See Live Display Function Buttons). When the Playback screen first appears, the last captured signal is displayed by default.

Selecting Captures for Playback

Select Event Button: When you select the button the Playback screen is overlaid with a dialogue box listing all the captured events stored in the buffer. Select the ones you wish to view and select the confirm button.

Restrictions During Playback

The following restrictions apply when using the Playback screen.

Because Playback is literally the replaying of previously recorded signals, it is only possible to make changes to the way in which the signals are displayed. It is not possible to alter anything that would affect the signals themselves. The following functions are therefore not available from the playback screen:

- Trigger configuration
- Output state control functions
- New capture button
- Clear graph

Live Display Sub-Tab

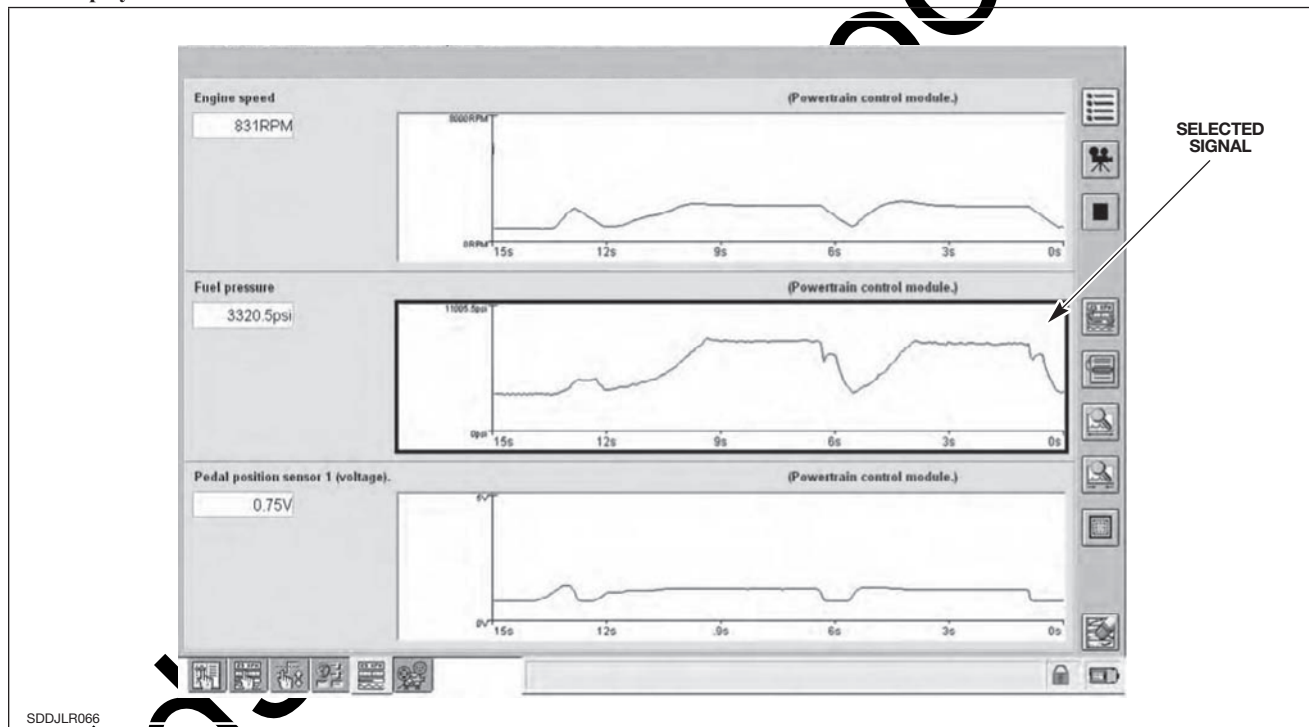
When you select the Live Display sub-tab, Datalogger verifies that the IDS is connected to the vehicle.

The Live Display screen and the Playback screen share certain features. The most obvious of these is the way in which signals are displayed.

Live Display Sub-Tab



Live Display Screen



You can select a signal simply by selecting the waveform (or bar graph/histogram) or signal name. If there are more signals, a vertical scroll bar will appear on the right of the screen to allow you to navigate between them.

Live Display Function Buttons

The Function Buttons associated with Live Display allow the user to capture and work on specific signals.

Select System Options

This button (which is always displayed regardless of sub-tab selection) allows you to print, road test and exit Datalogger.

Capture

This button only appears on the Live Display screen.

Play / Stop

This button only appears on the Live Display screen; it allows the Live Display to be stopped and re-started.

Zoom In

When this button is pressed, the horizontal (X-axis) divisions are enlarged by increments of 10%, 20%, 50%, 75%, 100%, 150%, 200%, 300% and 600%.

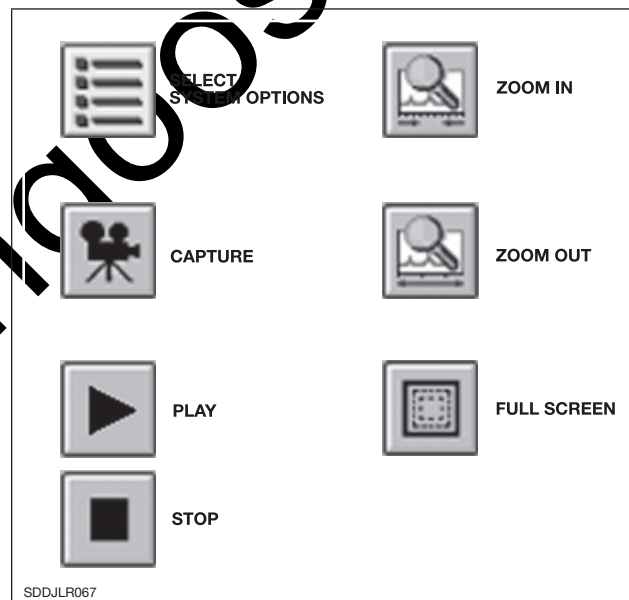
Zoom Out

When this button is pressed, the horizontal (X-axis) divisions are reduced by increments of 10%, 20%, 50%, 75%, 100%, 150%, 200%, 300% and 600%.

Full Screen

Press this button to display the selected signal full screen. Press the button a second time to restore the normal screen display.

NOTE: These are only some of the Function Buttons. To view the description of any Function Button, hold the stylus over the button for a few seconds; the description will be displayed in the secondary information bar at the bottom of the screen.



SIGNAL CAPTURE

Datalogger signals can be captured for review, either manually or by a Trigger Event.

Each session can store up to nine Datalogger capture events (recordings). When a new capture is made, all of the currently selected signals are stored for a duration defined by the values set in the capture buffer section of the Global Settings screen. Captures can be triggered automatically using the trigger settings, or manually by selecting the capture button. In either case, while capture is in progress a bar graph is displayed in the status panel.

Capture by Trigger

When a trigger event occurs, signals are automatically captured, provided trigger action is set to record, in Trigger Configuration. Refer to the Trigger section of this lesson for further information.

A capture event is completed either when its pre-set duration is reached, or when you select the stop button. In either case, you will be prompted to confirm whether the capture is to be saved, and if so to give it a name using the keypad. You can store up to nine captures. When this number is exceeded, further captures overwrite the oldest capture in the store.

The following restrictions apply to capturing:

- No distinction is made between captured signals and signals selected for display.
- An OSC signal can trigger a capture if you adjust its value to exceed its limit. This does not apply if a capture is already under way.
- Auto-trigger is disabled when all nine event buffers are filled.
- Captures can only be triggered by signals that are part of the selected signal set. This means, you cannot trigger on signal A and only capture signals B and C.

Global Settings

The Global Settings screen contains the configuration items common to all signals and provides a summary of the current configuration.

Pre- and Post-trigger settings can be changed as well as captured recorded time and capture rate.

Specifically, Global Settings allows you to set the duration of all captures within a session. The duration of Pre- and Post-triggers can be set so that additional data is recorded following a trigger event. This means that a session file can contain captures of varying duration.

Capture Duration

The capture duration is the recorded time of the signal data which is saved following either a trigger event, or following selection of the record button. The recorded data can then be viewed by the user.

The capture duration can be pre set in increments using the Fast, Medium, and Slow buttons as follows:

- Fast – 30 seconds
- Medium – 10 Minutes
- Slow – 12 Hours

Using Triggers

A trigger event is a point in time when a defined limit is reached or exceeded. A trigger can be set so that a Datalogger action is automatically carried out when the trigger event occurs. This action may be to record Datalogger signals or to sound an audible warning:

- **Record** All selected signals captured according to the current global settings (See Signal Capture). If you do not select a trigger action then the trigger becomes a set of limits on the display
- **Acoustic Warning** An audible warning sounds when the signal triggers. You can select Record on its own, Acoustic on its own or Record and Acoustic together, or select nothing.

If the trigger is set to 'Audible', then no recording occurs.

The progress indicator is activated while capture is in progress and navigation to other parts of the system is inhibited, although, scrolling and Stop Capture are enabled.

Setting a Trigger

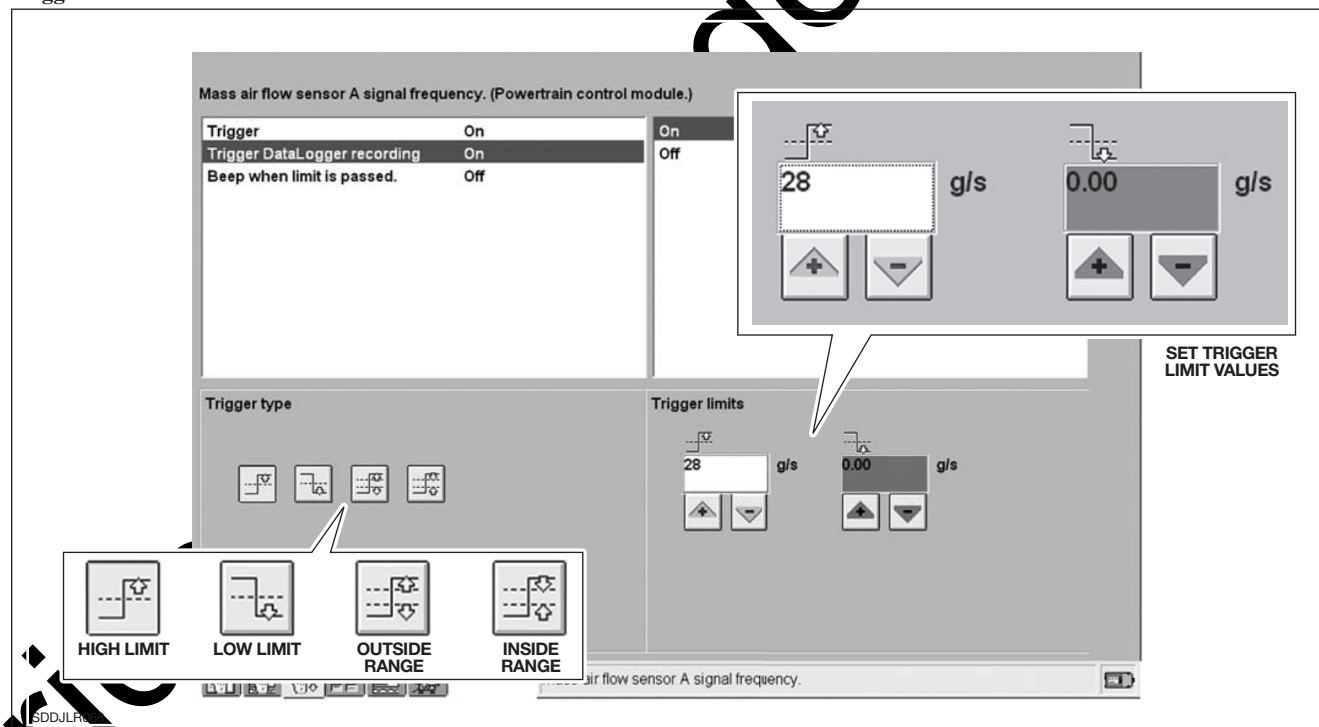
The signal which is required to be used as the trigger must first be selected. Following this, the trigger parameters must be set by selecting the trigger selection sub-tab.

Trigger Sub-Tab



Select the trigger screen by pressing the trigger sub-tab. The layout of the trigger screen is similar to the 'Signal Configuration' screen. The name of the signal selected as the trigger appears on the title bar at the top of the screen. The two panes immediately below the title bar indicate trigger action and trigger status. These will be On or Off. Set the trigger setting to On and select which action is to be taken when the trigger conditions are met.

Trigger Selection



Trigger Limits

A cluster of four buttons allow you to select the type of trigger. When either of these buttons is pressed, its background color changes to show it is live. At the same time, a thumbnail sketch indicating the chosen format appears in the pane above the cluster.

Trigger High Limit Select this to set a limit on a signal exceeding a pre-selected value

Trigger Low Limit Select this to set a limit on a signal dropping below a pre-selected value

Trigger Outside Range Select this to set a limit on a signal exceeding pre-selected upper and lower values

Trigger Inside Range Select this to set a limit on a signal dropping below pre-selected upper and lower values

Trigger Limit Values

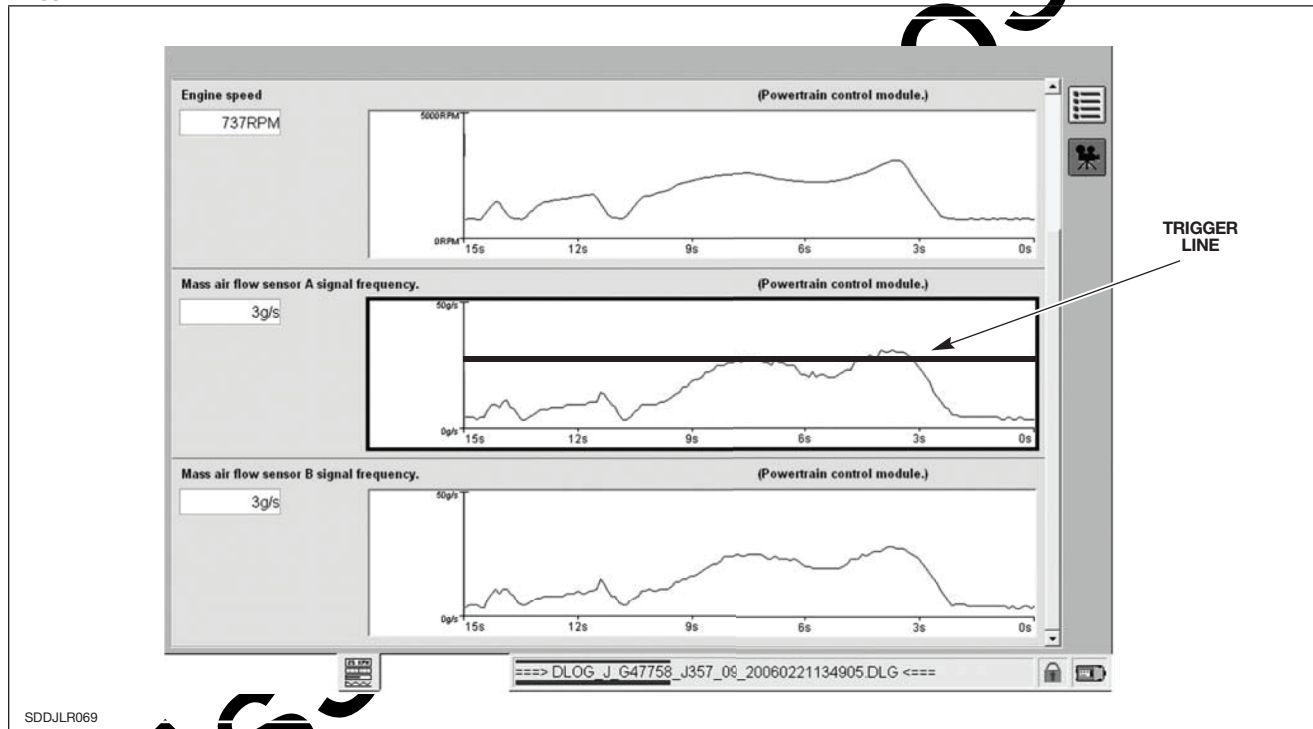
You can specify Trigger limit values by selecting the appropriate display pane and using the increment / decrement buttons. More accurate values can be set using the keypad. In the example shown here, the symbols for upper and lower trigger limits appear over their display panes. However, when only one trigger limit has been selected, the right-hand display pane is greyed out.

Run Datalogger

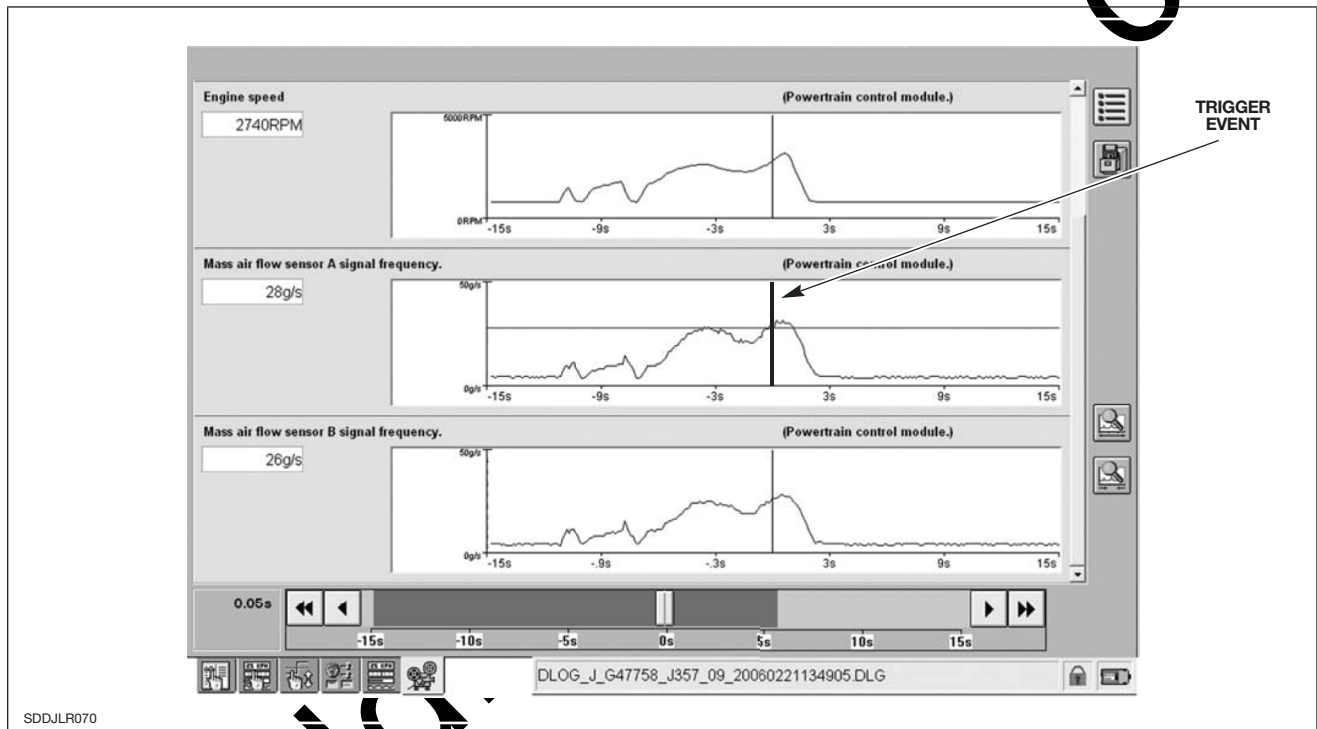
Once the trigger settings have been selected, navigate to the 'live display' screen and set Datalogger running.

The illustration below shows that the trigger has been set, shown here as a horizontal line. Once the trigger conditions have been met (the signal being displayed crosses this line) the capture will start. It can then be viewed when the capture period has completed.

Trigger Set



Capture Signal by Trigger Event



The trigger line is shown as a horizontal line, while the vertical line indicates the point at which the trigger event occurred.

The position of the vertical line can be moved to any point in time within the captured window, using the horizontal scroll buttons. The captured signal value is displayed in the window adjacent to the signal. The time, in seconds, when the trigger occurred, is shown on the horizontal axis. Zero represents the point at which the trigger event occurred.

Pre- and Post-Trigger Increments

NOTE: When a pop-up window appears in the screen display, the system only carries on working when one of the buttons offered in the window is selected. Any change to these settings after an event has been captured affects the size of succeeding buffers.

The pre- and post-trigger setting enables the proportion of the signal captured before and after a trigger event to be varied.

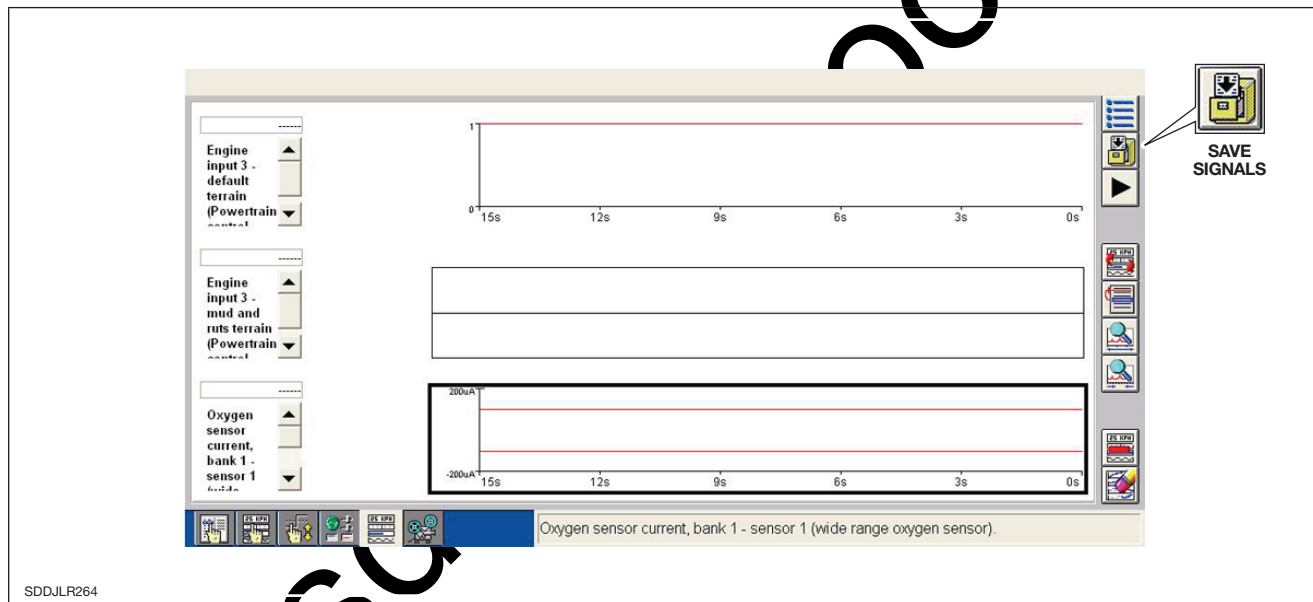
For example, if the capture duration is set to 20 seconds and the pre- and post-trigger setting is 20% and 80% respectively, then 4 seconds would be captured before the trigger event and 16 seconds would be captured after the event.

Datalogger Signal Grouping Feature

Datalogger Signal Grouping (DSG) allows users to create unique signal groups conforming to their individual needs. These signal groups can be saved and then recalled when needed.

Use the existing Datalogger interface to select and configure up to 16 Datalogger signals for viewing live.

You may also change the signal order, scaling, display type, triggers, etc., and create the arrangement of signals desired for a given purpose. Once the live display is set up to meet your needs, click on the new 'Save Signals' button (file cabinet with arrow down icon) in the button bar at the right side of the screen.



When the 'Save File' dialog window appears, give the file a name that will allow you to recognize the purpose of that signal group. Files will be saved in the required default location (the 'My Documents' folder in Windows®). DSG files can be identified by the 'DSG_' preceding the user-assigned name, as in this example:

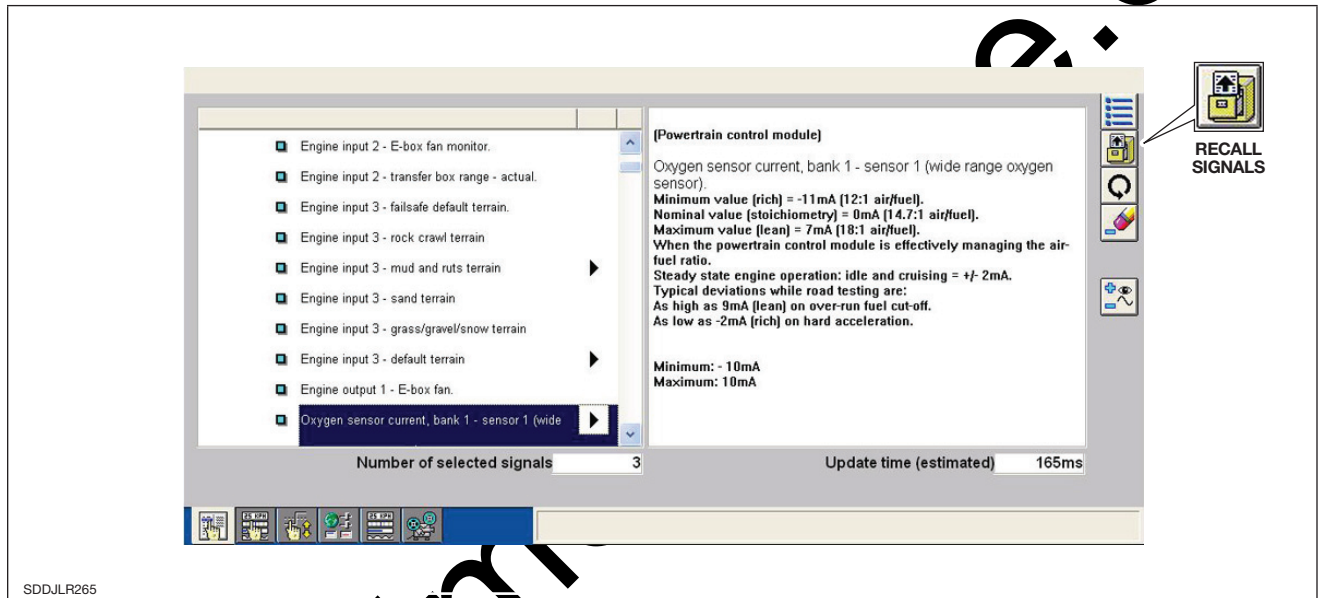
DSG_<user-assigned name>.XML

New DSG files are automatically assigned the model and model year range of the vehicle specifications for the current session.

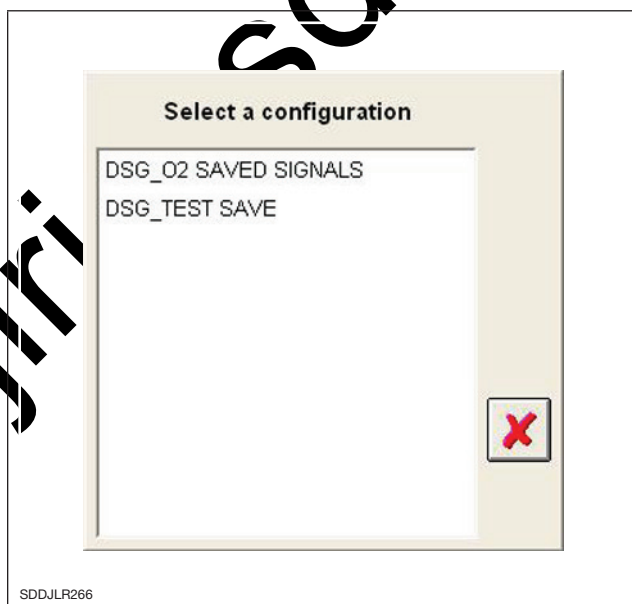
Users that wish to share DSG files across multiple IDS laptops can manually export/import DSG files from/to the My Documents folder. Note that if an IDS laptop requires a static restore and full system recovery, the DSG files must be manually backed-up and exported to a flash drive BEFORE executing a Phoenix Recovery/Static Restore process.

Recalling Existing Signal Groups

From the Signal Selection screen, click the new 'Recall Signals' button (file cabinet with arrow up icon) in the button bar.



This will pop-up a list of available DSG files for the current vehicle specification, as found by Datalogger in the default location.



Select the desired DSG file and the signal group will load into the live display viewer within Datalogger, with all the saved signal order, scaling, display type, triggers, etc. Click the Play button to begin viewing live data.

DSG File Associations and Vehicle Models

When new DSG files are created, Datalogger will automatically assign the currently specified vehicle model line. When DSG files are recalled, Datalogger will only present files compatible with the current vehicle specification.

Users must create unique DSG files for each vehicle variant for any given list of Datalogger signals. For example, to create a DSG file for oxygen sensors and fuel trim values to cover all Range Rover Sport vehicles, a total of 3 files would need to be created: one to cover 2006 – 2009 MY vehicles, another to cover 2010 – 2011 MY, and a third to cover 2012 MY.

If a DSG file is loaded that includes signals that are not applicable to the current vehicle because of optional equipment that is not present, those signals will not be added to the live display. Under these circumstances, SDD should alert the user to the conflict with a pop-up message window. The remaining applicable signals will be added to the live display.

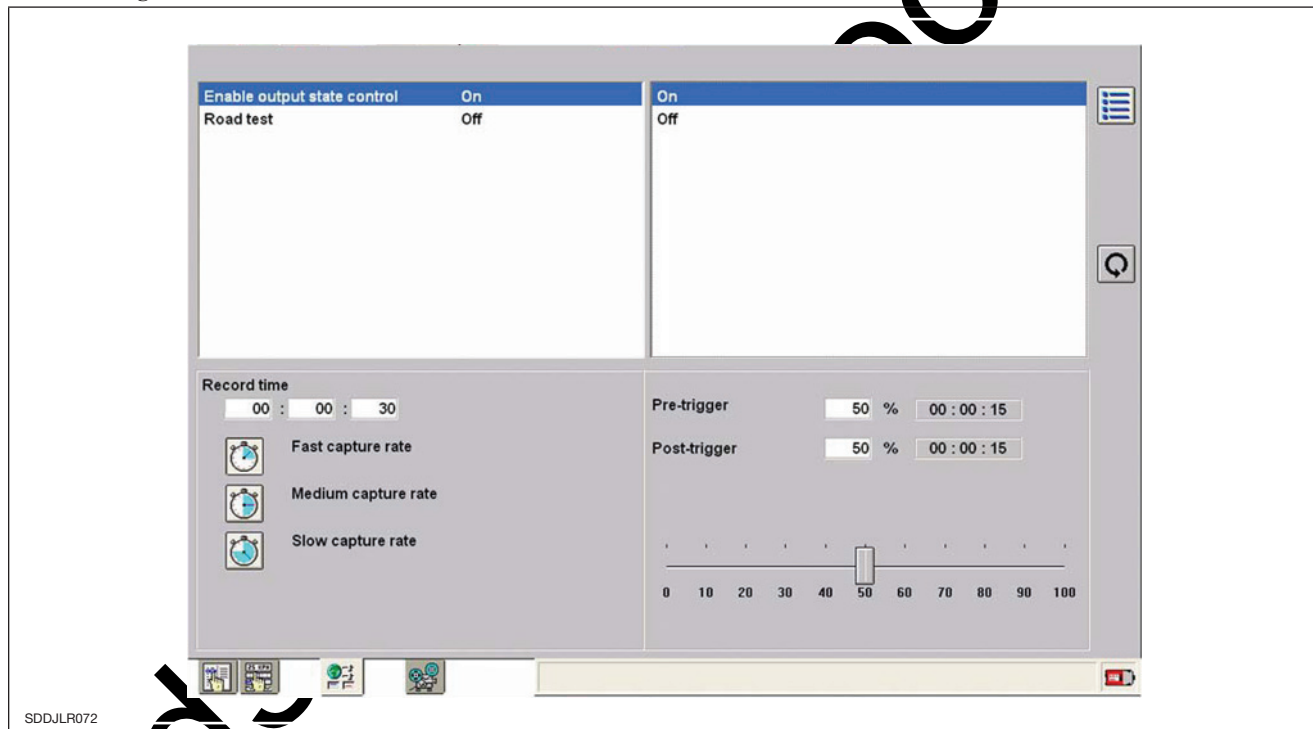
Road Test Mode

Road Test Mode allows the user to manually record a preselected Datalogger event while operating the vehicle safely. To access Road Test Mode, select the Global Configuration sub-tab.

Global Configuration Sub-Tab

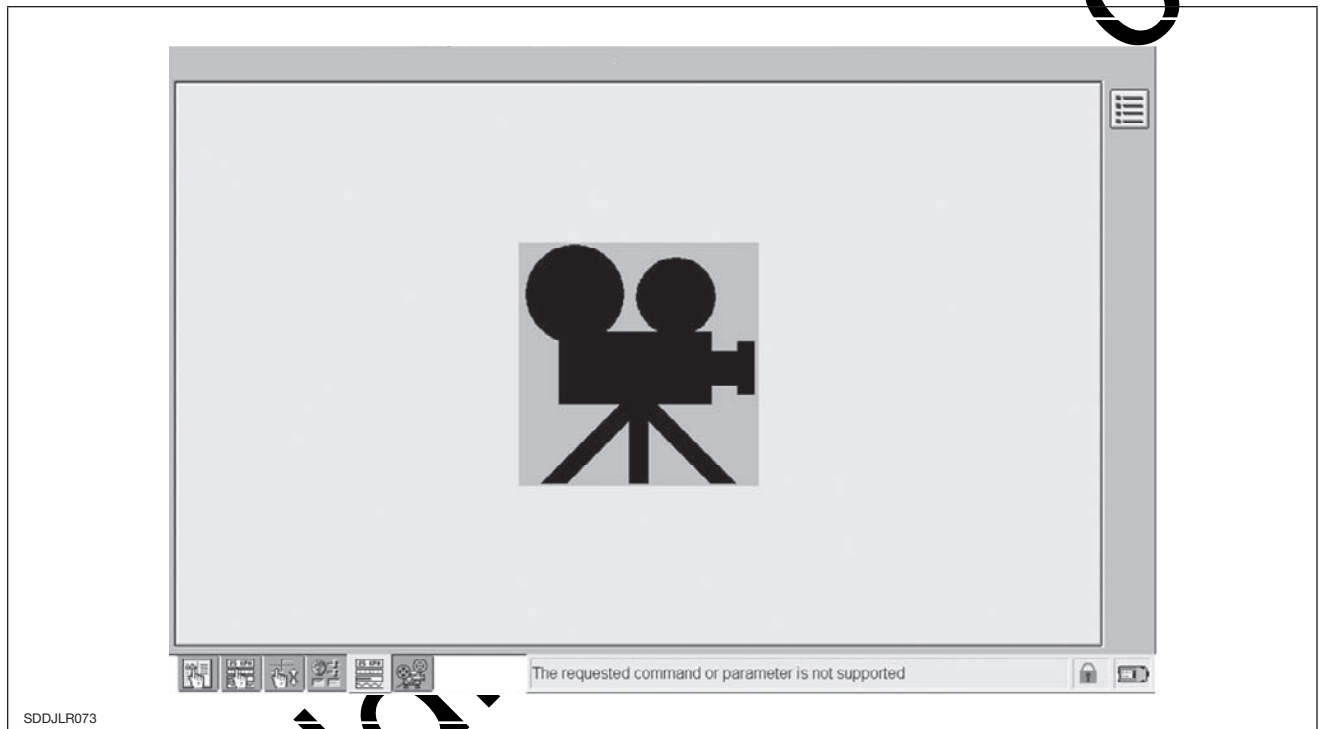


Global Settings Screen



Highlight 'Road test' in the left-hand pane of the display, then select On / Off in the right-hand pane.

Once Road Test is selected 'On', the following screen will appear.



IDS is now in Road Test Mode. Everything is blocked out on the screen and the user can easily touch anywhere on the screen to manually capture an event live while operating the vehicle safely. To stop the recording, the user will need to touch the screen again.

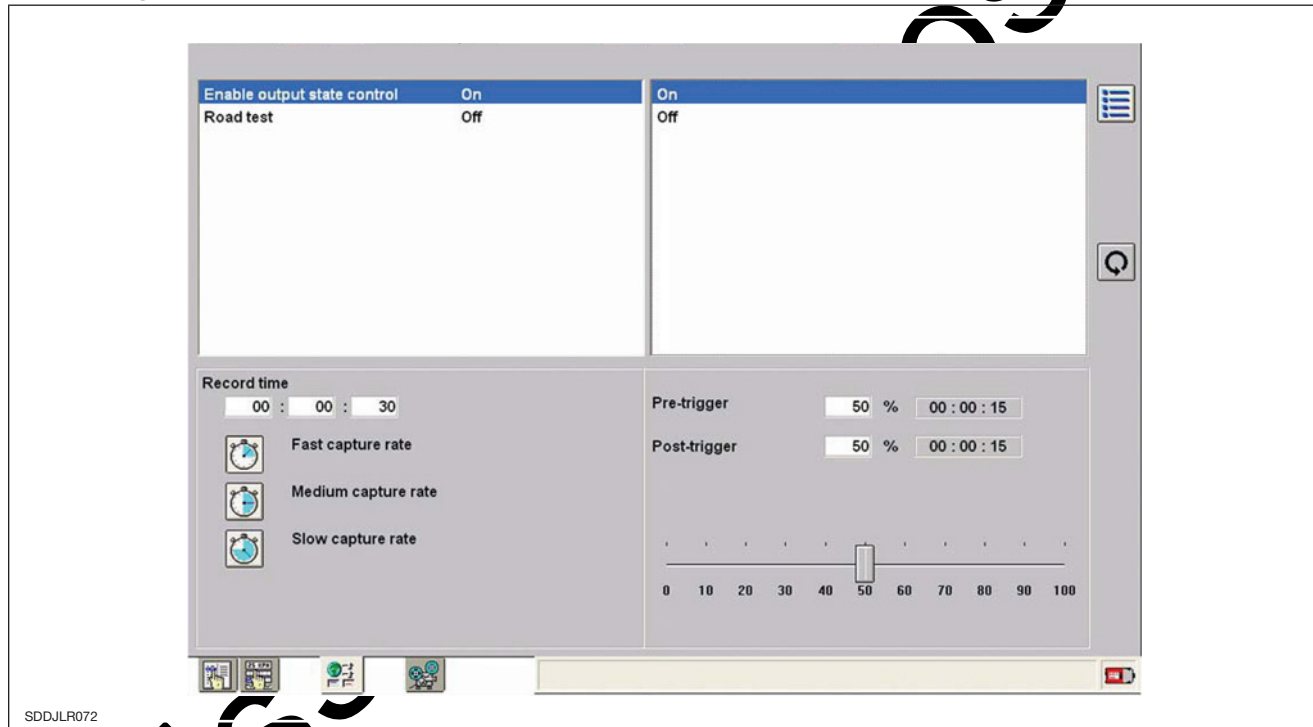
Up to nine recordings can be saved.

OUTPUT STATE CONTROL (OSC)

Output State Control (OSC) allows the user to control the voltage signal level output from a control module (CM) by directly assigning a value to it. This is only allowed for certain specific digital signals. Analog signals cannot be subjected to OSC.

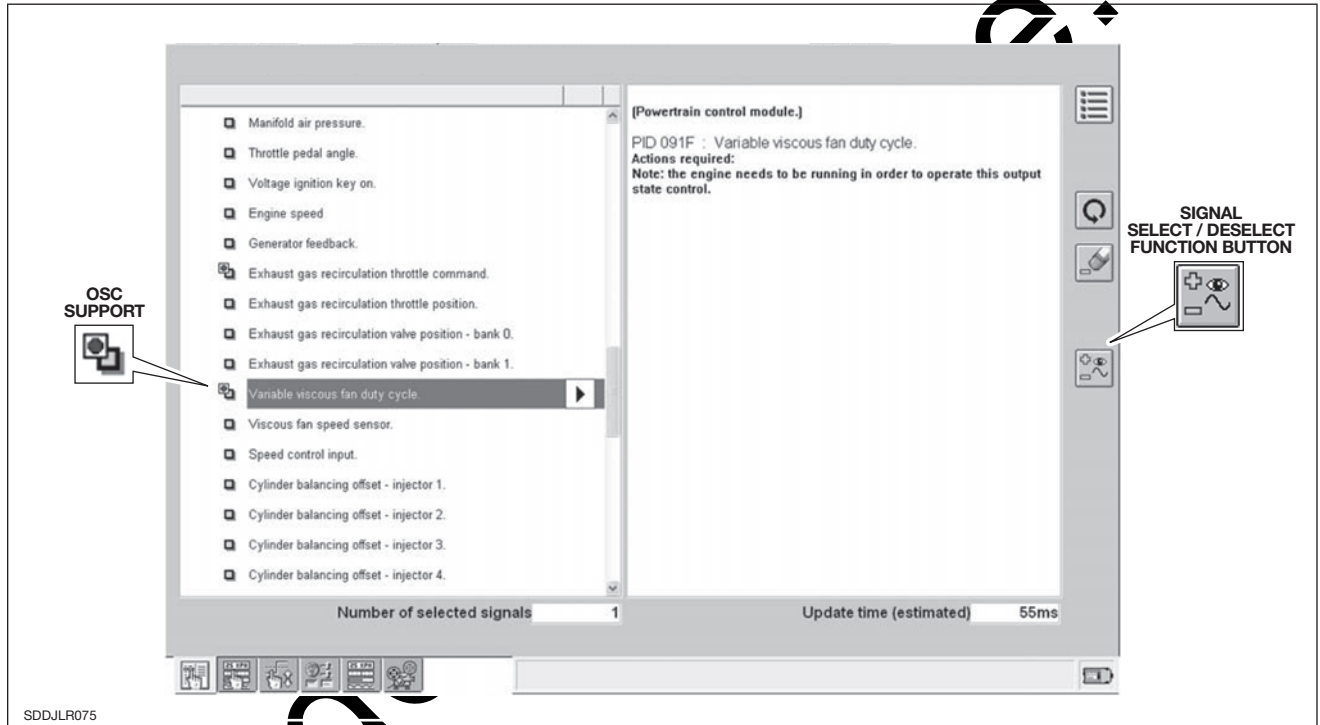
The user must have enabled OSC from the Global Configuration sub-tab for the function to be enabled.

Global Settings Screen



In the Datalogger signals list, signals that can have OSC applied to them are identified by an enhanced icon, as shown in the following illustration.

Signal Selection Screen – Output State Control Signals



NOTE: The OSC icon is only displayed if the desired system has been selected from the Content Model prior to selecting Datalogger.

Additional information regarding the conditions for operating the OSC signal, where applicable, is displayed in the right-hand pane of the screen once the signal has been highlighted.

In the illustration, 'Variable viscous fan duty cycle' is highlighted; in the right-hand pane, a note indicates that the engine needs to be running when applying the variable viscous fan duty cycle OSC signal.

NOTE: Highlighting a signal does not select it for Live Display. To select a signal, highlight it in the list, then operate the Select / De-select Signal function button. To select additional signals, repeat this sequence for each signal.

Controlling OSC

Once the signals have been selected, select the Datalogger Live Display sub-tab to display the signal(s). If more than one signal has been selected from the Datalogger menu, highlighting the displayed signal will cause the OSC function button to be displayed in the function button bar. Selecting the OSC function button will cause a display window and control buttons to appear on the screen to the left of the displayed signal.

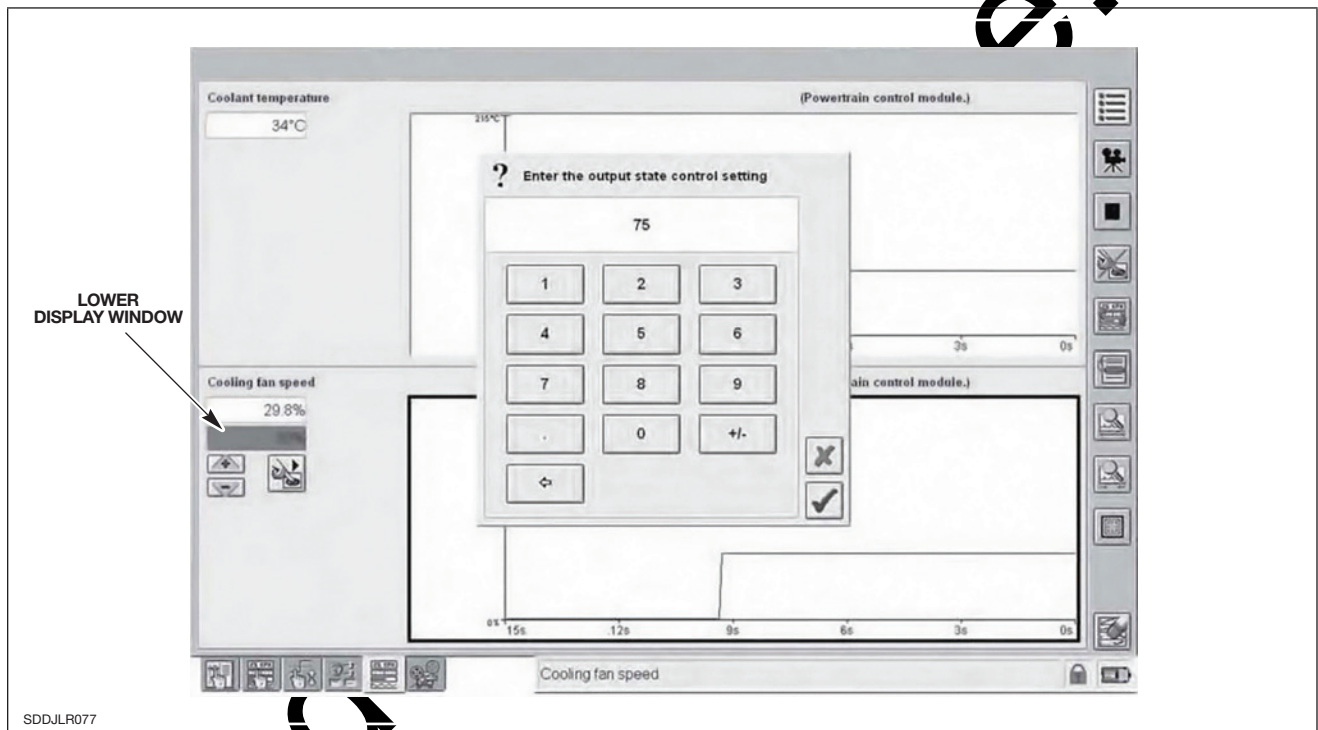


Selecting the appropriate increment buttons will change the OSC value. The current value is displayed in the upper window while the desired value is displayed in the lower window. Once the desired value has been achieved, selecting the Activate OSC function button will cause the control module to drive the selected component to the selected OSC value.

If OSC is enabled for the selected signal, the OSC increment/decrement buttons appear on the System Button bar at the right hand of the screen. The digital value of OSC is shown on the left of the signal display area. Unless you wish to alter a discrete OSC value (e.g. ON/OFF), use the buttons to change the value.

OSC Numerical Keypad

If the OSC value is to be changed by a large amount or changed quickly, operate the TSD over the lower display window. A numerical keypad will be displayed where the user can enter a desired value.



Confirm the selected value using the confirmation 'tick'.

Selecting the OSC activation button will apply the selected value to the component. Cooling Fan is now enabled at the desired speed. To stop OSC, select the Stop function button.

Output State Control – Cooling Fan Enabled



Output State Control Instruction Summary

1. Highlight subsystem from Content Model tab
2. Select Datalogger tab
3. Select desired signals to be monitored/controlled
4. Select Global Settings sub-tab
5. Select 'On' for 'Enable output state control'
6. Select Live Display sub-tab
7. Press the Play function button to begin the signal monitoring
8. Highlight the signal to be controlled by touching the signal display (a bold outline will surround the selected signal)
9. Press the Output State Control function button (Output State Control buttons will appear below the digital value box to the left of the highlighted signal)
10. Desired value can be selected two ways:
 - Incremental Up/Down buttons
 - Selecting the lower value display window will display a keypad that will allow numeric value entry
11. Press the Output State Control Play button to cause the entered value to be initiated.